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Cervical Vertebral Maturation Assessment using various Machine Learning techniques on Lateral cephalogram: A systematic literature review

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ABSTRACT

Importance: For the assessment of optimum treatment timing in dentofacial orthopedics, understanding the growth process is of paramount importance. The evaluation of skeletal maturity based on study of the morphology of the cervical vertebrae has been devised to minimize radiation exposure of a patient due to hand wrist radiography. Cervical vertebral maturation assessment (CVMA) predictions have been examined in the state-of-the-art machine learning techniques in the recent past which require more attention and validation by clinicians and practitioners.

Objectives: This paper aimed to answer the question "How are machine learning techniques being employed in studies concerning cervical vertebral maturation assessment using lateral cephalograms?"

Method: A systematic search through the available literature was performed for this work based upon the Population, Intervention, Comparison and Outcome (PICO) framework.

Data sources, study selection, data extraction and synthesis: The searches were performed in Ovid Medline, Embase, PubMed and Cochrane Central Register of Controlled Trials (CENTRAL) and Cochrane Database of Systematic Reviews (CDSR). A search of the grey literature was also performed in Google Scholar and OpenGrey. We also did a hand-searching in the Angle Orthodontist, Journal of Orthodontics and Craniofacial Research, Progress in Orthodontics, and the American Journal of Orthodontics and Dentofacial Orthopedics. References from the included articles were also searched.

Main outcome(s) and measure(s), results: A total of 25 papers which were assessed for full text, and 13 papers were included for the systematic review. The machine learning methods used were scrutinized according to their performance and comparison to human observers/experts. The accuracy of the models ranged between 60 and 90% or above, and satisfactory agreement and correlation with the human observers.

Conclusions and relevance: Machine learning models can be used for detection and classification of the cervical vertebrae maturation. In this systematic review (SR), the studies were summarized in terms of ML techniques applied, sample data, age range of sample and conventional method for CVMA, which showed that further studies with a uniform distribution of samples equally in stages of maturation and according to the gender is required for better training of the models in order to generalize the outputs for prolific use to target population.

1. Introduction

For the assessment of optimum treatment timing in dentofacial orthopedics, understanding the growth process is of utmost importance.[1] The skeletal disharmonies are remedied with growth modifications, which is referred to as dentofacial orthopedics. The best period for dentofacial orthopaedic treatment is during the pubertal growth period of an individual.[1,2]

Growth and development studies demonstrate that the timing, duration and speed of the pubertal growth spurt varies widely.[3,4] It is therefore necessary to evaluate the stage of skeletal maturity of an individual. A range of nearly trustworthy parameters has been used in determining skeletal maturity like chronological age, body weight and height, secondary sexual characteristics, dental maturity, skeletal maturity, etc.[5] Hand-wrist radiography is known to be used in

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Abbreviations

CVMA	Cervical vertebral maturation assessment
SR	Systematic review
PICO	Population, Intervention, Comparison and Outcome
DL	Deep Learning
AI	Artificial intelligence
ML	Machine learning
k-NN	k-nearest neighbors
NB	Naive Bayes
DT	Decision tree
ANN	Artificial neural networks
SVM	Support vector machine
RF	Random forest
LR	Logistic regression
CNN	Convolutional neural networks

establishment of skeletal maturation as the "gold standard" for the past years.[6,7]

A relatively new method for the evaluation of skeletal maturity based on study of cervical vertebra morphology was devised to minimize further radiation exposure of the orthodontic patient. The cervical vertebrae are clearly visible on the lateral cephalograms which are routinely used in orthodontics, thus providing an extra benefit of skeletal maturity assessment along with being a routine diagnostic aid.

The first seven parts of the spinal column are the cervical vertebrae. The vertebral growth of the superior and the inferior surfaces of every vertebra in the cartilaginous layer leads to changes in the size and shape of vertebrae. These morphological changes have been correlated to the increments in mandibular growth, thus aiding in indicating the ideal time for intervention prior to the peak in mandibular growth.

Several approaches have been described and their reliability has been proven in various published research in order to estimate skeletal maturity based on cervical vertebral analysis.[6–10]

The changes in the morphology of the second, third and fourth cervical vertebrae (C2, C3, C4) and in form of the body vertebrae from birth to maturation of the skeleton, are the basis of analyzing skeletal age by this method. Cervical vertebral maturation assessment (CVMA) is a notion strongly linked to skeletal maturity and an assessment of the pubertal growth spurt of the individual.

Machine learning (ML) is a sub-field of Artificial intelligence (AI) in which the machine learns how to carry out a task by mathematical analysis of the data sets provided rather than explicit programming of instructions.[11,12] This could be used in many fields of medical research, from genetics to diagnosis and treatment planning in various disorders. Machine learning is being increasingly used in orthodontics for various applications. Studies have documented the use of artificial intelligence (AI) and machine learning (ML) in orthodontics for tasks such as predicting orthodontic treatment length, determining skeletal classification, and predicting the need for orthodontic treatment or orthognathic surgery. Machine learning can also be used for analyzing various types of orthodontic records to provide an accurate and accelerated diagnosis. However, there are still challenges and limitations to the use of machine learning in orthodontics, and further research is needed to fully understand its potential and limitations.

CVMA predictions are being examined in the present state-of-the-art machine learning techniques.[11–13] In orthodontics, the precise CVMA prediction is extremely important. However, there are only a few encouraging results from some machine learning algorithms.[13] Various different studies have presented algorithms for CVMA prediction depending on specified points on the vertebrae, thus giving rise to varied algorithms. The performance of the ML learning depends on the training models used and the pre-specified points on the vertebrae taken into consideration for the CVMA prediction.

This review aimed to answer the question "How are different machine learning techniques being employed in studies concerning cervical vertebral maturation assessment using lateral cephalograms?" and to compare the various algorithms being used.

The data presentation is organized as follows: The methods which are used in accordance with recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement[14]; the results section collects and summaries data from the studies included in the review; the discussion section discusses results and presents concerns relating to validity threats; and the concluding statements and future directions of this research will finally be presented in the conclusion section.

2. Materials and methods

2.1. Literature search strategy

An electronic literature search was performed to identify suitable studies from Population, Intervention, Comparison and Outcome (PICO) framework,[15] in which 'Population' denotes the individuals undergoing CVMA, 'Intervention(s)' denote the machine learning or deep learning model using lateral cephalogram, 'Comparators' were the human experts who assessed the lateral cephalograms for CVMA by conventional methods and 'Outcomes' represents the agreement between ML model and conventional method of CVMA. This review is registered in PROSPERO with registration number CRD42021235048.

The literature search identified studies from Ovid Medline (U.S. National Library of Medicine), Embase (Elsevier), Cochrane Central Register of Controlled Trials (CENTRAL) and Cochrane Database of Systematic Reviews (CDSR) databases (with the necessary adaptations). The search included studies up to September 24th, 2022. The complete literature search strategy has been attached in Supplementary file-1. A search of the grey literature was also performed in Google Scholar and OpenGrey. Hand-searching in the Angle Orthodontist, Journal of Orthodontics and Craniofacial Research, Progress in Orthodontics, and in the American Journal of Orthodontics and Dentofacial Orthopedics was also done. References from the included articles were also searched and the results were saved electronically in EndNote Online (Clarivate Analytics), which was also used to remove the duplicates. The search string used for PubMed database search was as follows:

(((“age assessment” OR “age appraisal” OR “age diagnostics” OR “age estimate” OR “age estimation” OR “age determination” OR “age prediction” OR “age testing”) AND (“bone age measurement” OR “bone age assessment” OR “bone maturity” OR “bone development” OR “bone age testing” OR “bone age tests” OR “skeletal maturity” “skeletal maturation” OR “bone examination” OR “skeletal development” OR “developmental assessment” OR “bone age” OR “skeletal age” OR “growth zone” OR “cervical maturity index”)) AND (“x ray” OR “x-ray” OR “xray” OR “Radiography” OR “medical imaging”) AND (“machine learning” OR “unsupervised Machine Learning” OR “supervised Machine Learning” OR”Classification” OR”Regression” OR”Kernel” OR″Support vector machines” OR”Gaussian process” OR″Neural networks” OR″Logical learning” OR″relational learning” OR″Inductive logic” OR″Statistical relational” OR″probabilistic graphical model” OR″Maximum likelihood” OR″Maximum entropy” OR″Maximum a posteriori” OR″Mixture model” OR″Latent variable model” OR″Bayesian network” OR″linear model” OR″Perceptron algorithm” OR”Factorization” OR″Factor analysis” OR″Principal component analysis” OR″Canonical correlation” OR″Latent Dirichlet allocation” OR″Rule learning” OR″Instance-based” OR”Markov” OR″Stochastic game” OR″Learning latent representation” OR″Deep belief network” OR″Bio-inspired approach” OR″Artificial life” OR″Evolvable hardware” OR″Genetic algorithm” OR″Genetic programming” OR″Evolutionary robotic” OR″Generative and developmental approaches”)).

2.2. Inclusion criteria

Desired eligibility criteria included that the study should be published in English language, the study should address the research on age assessment using lateral cephalogram, should be a study concerning the building of models for the purpose of age assessment using at least one machine learning technique, should be a study in which the age assessment is analyzed on CVMI, should be a study of age assessment in living individuals, and lastly the outcome of models be compared to human observers.

Studies published in other languages, or which did not compare the outcome of ML models to assessment of CVM by human observers, or which did not use lateral cephalograms were excluded from this systematic review.

After the initial selection, the full texts were recovered for assessment of the methodological strength by two reviewers (PKC and SSR) for eligibility.

2.3. Study selection & data extraction

The search was carried out on September 14th, 2022, and the procedure for the same is detailed next.

From the total search of 320 articles, 72 articles were removed due to duplication. After the removal of duplicates, we screened 248 studies, assessing titles and abstracts, guided by the inclusion and exclusion criteria and further 223 records excluded. In total, 25 studies were selected for eligibility assessment. Among the 25 studies, 12 articles were excluded with reason after assessing their full texts.

2.3.1. Quality assessment

Risk of bias (ROB) in the included studies was assessed by two reviewers (PKC and SSR) by using Joanna Briggs Institute critical appraisal checklist for case-control studies observational cross – sectional studies.[16] (Table 1). The GRADE approach for nonrandomized studies was used for analysis of the strength-of-evidence for each of the outcomes. Disagreements between the reviewers were solved through a group reassessment of the article, until both reviewers were satisfied with the decision.

The final set of selected studies consisted of 13 papers.[13,17–26] Data extracted from the selected studies is presented in Table-2.

3. Results

3.1. Sample selection

Based on inclusion criteria, in this systematic review 25 articles have been selected for full text assessment (see Table 3). The article selection process has been described in the PRISMA flow chart which is shown in Fig. 1. The main reasons for exclusion of 12 articles from full text read have been described in Table- 3.[5,29–39] The initial screening of the title and abstract by the reviewers was in almost perfect agreement. The eligible studies were selected for data extraction and qualitative assessment. Statistics for inter rater agreement between two reviewers (PKC and SSR) was done using the Cohen's kappa statistic. Both the reviewers agreed to include 13 studies, and exclude 12 studies out of the selected 25 studies. A value of 1 was obtained, thus indicating perfect agreement between the raters.

3.2. Sample distribution

The data distribution shows sample sizes ranging from 300 up to 1870 distributed among various countries mostly Asian countries including Korea,[20,23,26] China,[21,25] Iran[19] and Turkey,[13,17,24,27,28] and one study from France.[18] Another study[22] was performed retrospectively on the database of growth studies of Michigan, IOWA and Bolton Brush, thus having very few data on the Caucasian population.

Table 1
Details of the risk-of-bias of the included studies according to the Joanna Briggs Institute's critical appraisal checklist.

Study	Were the criteria for inclusion in the sample clearly defined?	Were the study subjects and the setting described in detail?	Was the exposure measured in a valid and reliable way?	Were objective, standard criteria used for measurement of the condition?	Were confounding factors identified?	Were strategies to deal with confounding factor stated?	Were the outcome measured in a valid and reliable way?	Was appropriate statistical analysis used?	Overall Risk of Bias
Mohammad-Rahimi H, 2022	Unclear	No	Unclear	Yes	No	No	Yes	Yes	37.5% High
Seo et al, 2022	Yes	No	Unclear	Yes	Yes	No	Yes	Yes	62.5% Moderate
Liao N et al, 2022	No	No	Unclear	Yes	Yes	No	Yes	Yes	50% Moderate
Atici et al, 2022	No	No	Unclear	Yes	Yes	No	Yes	Yes	50% Moderate
Kim et al, 2021	No	No	Unclear	Yes	Yes	No	Yes	Yes	50% Moderate
Kok et al, 2021	Yes	Yes	Unclear	Yes	Yes	No	Yes	Yes	75% Low
Zhou et al, 2021	Yes	Yes	Unclear	Yes	Yes	No	Yes	Yes	75% Low
Seo and Hwang, 2021	Yes	No	Unclear	Yes	Yes	No	Yes	Yes	62.5% Moderate
Kok H et al, 2020	Yes	Yes	Unclear	Yes	Yes	No	Yes	Yes	75% Low
Amasya H et al, 2020	Yes	Yes	Unclear	Yes	Yes	No	Yes	Yes	75% Low
Amasya H et al, 2020	Yes	Yes	Unclear	Yes	No	No	Yes	Yes	62.5% Moderate
Kök H et al, 2019	Yes	Yes	Unclear	Yes	No	No	Yes	Yes	62.5% Moderate
Makaremi et al, 2019	Yes	No	Unclear	Yes	Yes	No	Yes	Yes	62.5% Moderate

Table 2
Characteristics of studies included in the systematic review.

Sr. No	Author, Year	Country	Sample size	Age	Training/ Testing ratio	Conventional Method used	Machine learning method	Human comparison	Outcome
1	**Mohammad-Rahimi H, 2022**	Iran	890; ROI cropped	Not mentioned	70–30	McNamara & Franchi	Supervised learning; Transfer learning models; pre-trained and newly trained for 2 data sets.	2 Orthodontists	ResNet101 showed best performance. Six-class CVM diagnosis in ResNet101 model showed validation and test accuracy of 62.63% and 61.62%, respectively. With three class classification, model's validation and test accuracy were 75.76% and 82.83%, respectively. But all models showed lesser accuracy than human observers.
2	**Seo et al, 2022**	Korea	900; ROI cropped	4–18 years	80–20	Baccetti et al.	Supervised learning; Convolutional neural network; 1 pre-trained model and 1 newly trained model	Not mentioned	The segmentation model possessed average accuracy, intersection over union, and mean boundary F1 scores of 0.956, 0.913, and 0.895, respectively, for the segmentation of cervical vertebrae from lateral cephalogram. The regression model for estimating bone age from segmented cervical vertebrae images yielded average mean absolute error and root mean squared error values of 0.300 and 0.390 years, respectively. Strong correlation was also observed between the real and predicted bone age obtained using the automatic and manual ROI methods with the coefficients of determination being 0.983 and 0.934, respectively.
3	**Liao N et al, 2022**	China	900; sub-set cropped	7–25 years	80–20	McNamara & Franchi	Unsupervised learning; Label distribution learning, deep learning; newly trained model	3 experts in Orthodontics and Radiology	iCVM on CVM-900 showed 69.1% accuracy and on CVM-900-Subset showed 84.26% accuracy. The QWK values of the proposed iCVM on CVM-900 and CVM-900-Subset reach 0.947 and 0.963 respectively, gaining an almost perfect agreement between the predicted rating and the actual rating. The MAE value on the clean subset CVM-900-Subset is 0.16, which means on average there is a gap of 0.16 stage between the predicted stage and the ground-truth.
4	**Atici et al, 2022**	*Study performed on IOWA, Michigan and Bolton Brush*	1018	4–29 years	70–30	McNamara and Franchi & Baccetti et al.	Supervised learning; Convolutional Neural Network, residual learning; pre- trained and newly trained model	1 Orthodontist	The proposed method yielded 75.11% accuracy for six classes (CS1-CS6). It achieved a higher 84.63% validation accuracy in 5-stage

(*continued on next page*)

Table 2 (*continued*)

Sr. No	Author, Year	Country	Sample size	Age	Training/ Testing ratio	Conventional Method used	Machine learning method	Human comparison	Outcome
		Facial Growth Studies							classification (CVMS I-CVMS V). The model performed better on 5-stage classification than on 6-stage classification.
5	**Kim et al, 2021**	Korea	600	6–18 years	80–20	McNamara and Franchi	Supervised learning; Convolutional Neural Network; pre-trained and newly trained	2 specialists	Model accuracy was evaluated using the Epsilon index, allowing one- or two-stage errors. Model_3 showed the highest accuracy in terms of Epsilon, which is a significant improvement over models 1 and 2. The accuracy of model_3 was 9.17% higher than that of model 1 and 3.34% better than that of model 2. More than 93% accuracy was achieved for Epsilon (1).
6	**Kok et al, 2021**	Turkey	360	8–17 years	80–20 and 70–30	Hassel and Farman	Supervised learning; Artificial neural network model (NNM) and Naïve Bayes model (NBM); newly trained models	1 Orthodontist, twice at 15 day interval	The highest determination success rate was obtained in NNM 3 (0.95) and the lowest in NBM 4 (0.50). The determination success of NBM 1 and NBM 3 was almost similar (0.60). The success of NNM 2 did not differ much from that of NNM 1 (0.94). The determination success of stage 5 was relatively lower than the others in NNM 1 and NNM 2 (0.83). The NNMs were more successful than the NBMs in our developed models. It is important to determine the effective ratio and/or measurements that will be useful for differentiation.
7	**Zhou et al, 2021**	China	1080	6–22 years	90–10	Baccetti et al.	Supervised learning; Convolutional neural network; newly trained models	2 examiners; Disagreements resolved by third examiner	The mean labelling error between human examiners was 0.48 mm. The mean labelling error between AI and human examiners was 0.36 mm. In general, the agreement between AI results and the gold standard was good, with the intraclass correlation coefficient (ICC) value being up to 98%. Moreover, the accuracy of CVM staging was 71%. In terms of F1 score, CS6 stage (85%) ranked the highest accuracy.
8	**Seo and Hwang, 2021**	Korea	600 (Only 100 cephs read by human)	6–19 years	80–20	Baccetti et al.	Supervised learning; Convolutional Neural network; pre-trained and newly trained models	Not mentioned	All deep learning models demonstrated more than 90% accuracy, with Inception-ResNet-v2 performing the best, relatively. In addition, visualizing each deep learning model using Grad CAM led to a primary focus on the

(*continued on next page*)

Table 2 (*continued*)

Sr. No	Author, Year	Country	Sample size	Age	Training/ Testing ratio	Conventional Method used	Machine learning method	Human comparison	Outcome
									cervical vertebrae and surrounding structures. The use of these deep learning models in clinical practice will facilitatedental practitioners in making accurate diagnoses and treatment plans.
9	**Kok H et al, 2020**	Turkey	419	8–17 years	70–30	Hassel and Farman	Supervised learning; Artificial Neural Network; newly trained models	1 Orthodontist, twice at 20 day interval	Significantly positive correlations between hand-wrist maturation level, CVS and ages. ANN-7 model accuracy value was 0.9427. Highest model accuracy 0.8687 with least linear measurements was obtained by drawing 13 linear measurements, using vertical measurements and indents. The growth development periods and gender were determined from CVM using ANN successfully.
10	**Amasya H et al, 2020**	Turkey	647	10–30 years	80–20	Baccetti et al.	Supervised learning; Logistic regression (LR) and Artificial neural network models (NNMs); newly trained models	Two observers	Percentage agreement between the model and the visual analysis of the researcher was 86.93%, which was the highest among all models tested
11	**Amasya H et al, 2020**	Turkey	647	10–30 years	80–20	Baccetti et al.	Supervised learning; Artificial Neural Network; pre-trained and newly trained models	2 radiologists	Intraobserver agreement ranges were as follows: $w\kappa = 0.92–0.98$, $c\kappa = 0.65–0.85$, and 70.8%–87.5%. Interobserver agreement ranges were as follows: $w\kappa = 0.76–0.92$, $c\kappa = 0.4–0.65$, and 50%–72.2%. Agreement between the ANN model and observers 1, 2, 3, and 4 were as follows: $w\kappa = 0.85$ ($c\kappa = 0.52$, 59.7%), $w\kappa = 0.8$ ($c\kappa = 0.4$, 50%), $w\kappa = 0.87$ ($c\kappa = 0.55$, 62.5%), and $w\kappa = 0.91$ ($c\kappa = 0.53$, 61.1%), respectively ($P < 0.001$). An average of 58.3% agreement was observed between the ANN model and the human observers.
12	**Kök H et al**, 2019	Turkey	300	8–17 years	80–20	Hassel & Farman	Supervised learning; k-nearest neighbors (k-NN), Naive Bayes (NB), decision tree, ANN, support vector machine (SVM), RF and logistic regression; pre-trained models	1 Orthodontist, evaluated twice at two different time points	The neural network model had the second highest accuracy values for determining individual stages, except the fifth stage, but was the most stable among all the algorithms tested.
13	**Makaremi et al, 2019**	France	1870	Not mentioned	80–20	McNamara and Franchi	Supervised learning; Convolutional neural network; pre-trained and newly trained models	1 radiological technician	The results show the performances of the proposed method in different contexts with different number of images for training, evaluation and testing and different pre-processing of these

(*continued on next page*)

Table 2 (*continued*)

Sr. No	Author, Year	Country	Sample size	Age	Training/ Testing ratio	Conventional Method used	Machine learning method	Human comparison	Outcome
									images. The proposed model and method are validated by cross validation.

Table- 3
Reason of article excluded after full text read.

S. No.	Study	Reason for exclusion
1	Xie et al. (29)	No use of lateral cephalogram; directly sagittal slice taken.
2	Feng et al. (30)	Lateral cephalogram not used to develop the model; sagittal slice of CBCT taken.
3	Kim et al. (31)	Use of CVM to predict hand-wrist SMI
4	Aguiar et al. (32)	Exclusive use of Caldas formula in regression model for CVMI assessment
5	Banumathi et al. (33)	Exclusive use of landmark on other craniofacial regions other than cervical vertebrae
6	Litsas et al. (34)	Exclusive use of Dental calcification stages for determination of CVMA
7	Moon et al. (35)	Exclusive use of landmarks on the other craniofacial regions other than cervical vertebrae
8	Dzemidzic V et al. (5)	Computer-based assessments with manual interaction. Software-based assistance only with formulas but not intelligence
9	Santiago RC et al. (36)	Computer-based assessments with manual interaction. Software-based assistance only with formulas but not intelligence
10	Chen L et al. (37)	Computer-based assessments with manual interaction. Software-based assistance only with formulas but not intelligence.
11	Lindner C et al. (38)	Exclusive use of landmark on the other craniofacial regions other than cervical vertebrae
12	Bapista et al. (39)	Computer-based assessments with manual interaction. Software-based assistance only with formulas but not intelligence

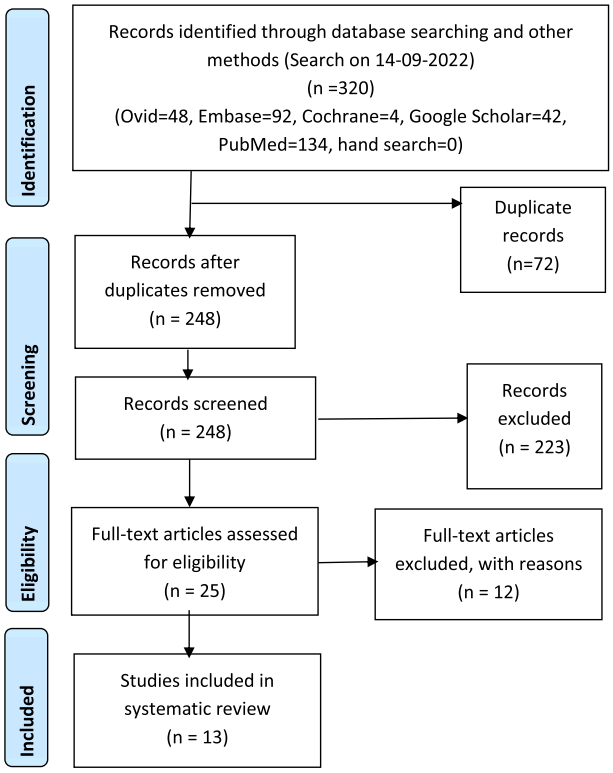


Figure: 1. Statement flowchart of article selection process
(From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021; 372:n71. https://doi.org/10.1136/bmj.n71) (For more information, visit: http://www.prisma-statement.org/).

3.3. Method of CVM assessment

Among the conventional methods used for cervical assessment, most of the studies[13,20,25,26,28] have used Baccetti et al. method,[1] studies by Kok[17,24,27] have used Hassel and Farman method[8] for cervical vertebrae assessment, and McNamara and Franchi method has been used by a few studies.[18,19,21,23] Only one study has compared between the Baccetti method and the McNamara and Franchi method.[22] The experience of the human observers who classified the lateral cephalograms, and their field of expertise also varied. Few studies classified the cephalograms twice at a specific interval,[17,24,27] or involved more than one observer to avoid error and improve the standards of the study.[13,19,21,23,25,28]

3.4. Training models

Six studies[13,19,20,22,23,26] proposed newly trained models and further compared their performance with pre-trained models. A few other studies proposed newly trained models,[21,24,25,27,28] but it was difficult to evaluate their performance as compared to other pre-existing models. Only one study[21] used unsupervised learning, which implemented label distribution learning and deep learning models.

3.5. Performance of machine learning models

Only two studies[18,22] made use of directional filters which serve as an edge detector and help in improving the performance of the model. These two studies have shown superior performance since the first derivatives (or slopes) of an image are most evident when a large change occurs between adjacent pixel values. Thus, the directional filters were designed for the morphological changes in the shape of the cervical vertebrae in these studies. The implementation of unsupervised learning in a study[21] was based on label distribution learning with loss function, which helped in assessing on how many occasions was their model able to correctly predict the CVMA and when it couldn't.

All studies with cropped region of interest (ROI) showed good correlation except for the study by Kim et al.[23] Only one study by Kok et al.[24] did not implement cropping of the ROI of the cervical vertebrae. Two studies[20,24] also used the neural network model for gender determination along with skeletal age determination. Overall, the accuracy of the models of the various studies ranged between 60 and 90% or above, and satisfactory agreement and correlation with the human observers was achieved, except in the study by Mohammad et al.[18] which claimed more accuracy of the human observers than that of the models.

4. Discussion

Lateral cephalograms have been utilized in orthodontics for diagnosis and treatment planning, and skeletal age assessment using cervical vertebrae. Therefore, the need of an extra hand-wrist radiograph is eliminated to evaluate skeletal maturity. This review examined the use

of machine learning models in CVM assessment, including 13 articles in total.

Machine learning[29] was first conceived from the mathematical modeling of neural networks, when logician Walter Pitts and neuroscientist Warren McCulloch in 1943, attempted to mathematically map out thought processes and decision making in human cognition. In 1950, Alan Turing proposed the Turing Test, which became the litmus test for which machines were deemed "intelligent" or "unintelligent." In early 1960s, several groups were engaged in the design and testing of learning recognition systems. Soon, a summer research program at Dartmouth College became the official birthplace of AI.

From this point on, "intelligent" machine learning algorithms and computer programs started to appear, to do everything from using speech recognition to learning to pronounce words the way a baby would learn to aiding in diagnosis in medical sciences.

In Orthodontics, conventional cervical vertebrae assessment methods are time consuming, and also are subjective to a certain extent. A machine learning model could be time saving and also aid in removing biased opinions about the identification of the stage of cervical maturity.[41,42] Neural networks are a subset of machine learning implemented in the healthcare field. They are comprised of node layers, containing an input layer which is the image/dimensions of the cervical vertebrae, one or more hidden layers, and an output layer which is the predicted CVMA stage with the highest probability. Each node connects to another and has an associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network. This process of machine learning avoids the need of any human intervention for assessment.

Various tasks can be accomplished with different ML techniques. Choosing the appropriate technique is essential, keeping in view the expected outcome or prediction from the model. Classification algorithms learn from labelled examples (existing CVMA method) and produce a model for categorization of the unlabeled examples (the samples fed to the model for prediction of the CVM). Classification and regression techniques are associated with supervised learning, whereas clustering is associated with unsupervised learning and unlabeled data.[43,44] Several classification techniques can be used in ML (e.g., artificial neural network (ANN), logistic regression (LR), decision tree (DT), random forest (RF) and support vector machine (SVM)).

This systematic review of the literature has been performed to understand the use and application of machine learning models as compared to the conventional CVMA methods. This work was done by answering a question developed with PICO strategy.

The included studies in the review have used the methods by Hassel and Farman[8] and Baccetti et al.[1] and McNamara and Franchi's method, which are both subjective methods since they depend on visual assessment of the shape vertebrae C3 and C4, and the presence of a concavity on the inferior aspect of C2, C3 and C4 vertebral bodies. Since these methods are the most used in literature, it considerably increased the probability of acquiring relevant literature on the subject.

Thirteen studies were included for this systematic review. Other studies which were excluded used different software for CVMA using a computer-based assessment with manual interaction, hence were not included in this review.[5,45,46] Two studies[30,31] were excluded since they used sagittal slice derived from either a CT scan or a CBCT scan visualizing the cervical vertebrae, and didn't make use of the lateral cephalograms. Three other studies[33,34,36,39] were excluded since they used machine learning for landmark identification in the entire craniofacial region, and not only in the cervical vertebral region for CVM assessment. Another study[35] used dental calcification stages to predict skeletal maturity of the cervical vertebrae, and was thus excluded. Four other studies[5,37,38,40] made use of computer-based assessments with manual interaction and not only machine learning for the skeletal maturity assessment and thus were not included in the systematic review.

The data samples from the studies are further divided into subsets which aids in validation of the proposed model.[46] The training-testing ratio also reflects on the validity of the model, since most of the studies showed a healthy ratio of 4:1. However, all studies had a variable age group and samples were not selected according to the CVMI status, thus unequal number of samples belonged to each CVMI group. An ideal sample would include equal samples in each CVMI group which can help in further training of the machine learning models for that particular CVMI stage.

Most of the studies using AI for determining the CVM stage are based on supervised learning since supervised learning models tend to be more accurate than unsupervised learning models. This is because they require upfront human intervention to label the appropriate CVM stages initially. The labelling is based on the pre-existing methods[1] of using two dimensional cephalograms, which now needs revision according to the current studies using 3D diagnostic technologies such as CBCT scans.

Studies by Amasya H et al.[13,28] used Baccetti's method[1] for CVMA and developed a new software named clinical decision support system (CDSS) for manual extraction of image features of digital lateral cephalograms with marking of the anatomical landmarks. They compared the conventional analysis with five different classifier models developed using AI techniques (LR, DT, RF, ANN and SVM). Their first study[13] was the first study to compare the various machine learning models, to determine which suits best for prediction of CVMA, and concluded that the ANN model performed the best.

Another set of studies included in this review were by Kok et al.[17,24,27] These studies used 300–400 cephalometric radiographs and used 7 different artificial intelligence algorithms. These algorithms are k-nearest neighbors (k-NN), Naive Bayes (NB), DT, ANN, SVM, RF, and LR algorithms. As a result of their initial study,[17] they found that overall, the ANN algorithm gave the most consistently stable prediction, however the area under curve (AUC) evaluation showed that the ANN was weaker in determining the CVS3 and the CVS5 stages. Further studies by the same author used ANN models to correlate bone age based on hand wrist radiography to CVMA, and further added their input by adding linear measurements for CVMA prediction.[24,27]

The pilot study by Mohammad-Rahimi[19] used CNN which power image recognition and computer vision tasks. Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs. The ResNet algorithm showed the best performance among the various algorithms they tested to develop the AI model. CNN models were also used in series of studies by Seo et al.[20,26] which made particular use of stepwise regression modeling by adding input variables to improve the model performance. Reliability of using CNN for CVMA as compared to human examiners was assessed in another study,[25] showing 97% reliability.

Results of most of the studies have been expressed in confusion matrix. For different ML model success using confusion matrices, sensitivity (recall), precision, area under the curve (AUC), and F1 criterion have been used.

The results of the systematic review showed that every article selected presented positive agreement between the conventional and machine learning method for assessment methods of skeletal maturation.

5. Limitations

This systematic review has limitations of not including any AI expert of engineering background. Included studies show agreement between ML method and conventional method of CVMA, although data distribution is not uniform through all stages of CVM maturation. An attempt to conduct a meta-analysis was made, although the statistics were heterogenous in nature.

6. Future perspectives

In future, ML could be used for CVMA and save the time of clinician or researcher. We suggest to perform a meta-analysis in future with studies including homogenous statistics. Another prospect is the current trend of three-dimensional (3D) diagnostic records such as Cone Beam Computed Tomography, Computed Tomography and Magnetic Resonance Imaging, which are being used for landmark identification as well as in performing analysis. Implementation of deep learning models with 3D input data may prove to be more accurate than current CVM assessment from lateral cephalograms.

7. Conclusion

This systematic review summarizes the ML techniques applied, sample data, age range of sample and conventional method for CVMA. Keeping the research question in mind, and based on the results found, it is innocuous to confirm that ML model have positive agreement with conventional CVMA. However, the studies included in this review have non-homogenous data and statistics, which suggests that elaborations of more specific study designs for clarification are required.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jobcr.2023.08.005.

References

1 Baccetti T, Franchi L, McNamara JA. The cervical vertebral maturation (CVM) method for the assessment of optimal treatment timing in dentofacial orthopedics. *Semin Orthod*. 2005;11:119–129.
2 Faltin K, Faltin R, Baccetti T, Franchi L, Ghiozzi B, McNamara JA. Long-term effectiveness and treatment timing for bionator therapy. *Angle Orthod*. 2003;73(3):221–230.
3 Silventoinen K, Haukka J, Dunkel L, Tynes P, Rasmussen F. Genetics of pubertal timing and its associations with relative weight in childhood and adult height: the Swedish young male twins study. *Pediatrics*. 2008;121:e885–e891.
4 Parent AS, Teilmann G, Juul A, Skakkebaek NE, Toppari J, Bourguignon JP. The timing of normal puberty and the age limits of sexual precocity: variations around the world, secular trends, and changes after migration. *Endocr Rev*. 2003 Oct;24(5):668–693. https://doi.org/10.1210/er.2002-0019. PMID: 14570750.
5 Dzemidzic V, Sokic E, Tiro A, Nakas E. Computer based assessment of cervical vertebral maturation stages using digital lateral cephalograms. *Acta Inf Med*. 2015;23(6):364–368.
6 Fishman LS. Radiographic evaluation of skeletal maturation. A clinically oriented method based on hand-wrist films. *Angle Orthod*. 1982;52:88–112.
7 Bjork A, Helm S. Prediction of the age of maximum pubertal growth in body height. *Angle Orthod*. 1967;37(2):134–143.
8 Hassel B, Farman AG. Skeletal maturation evaluation using cervical vertebrae. *Am J Orthod Dentofacial Orthop*. 1995;107:58–66.
9 Flores-Mir C, Burgess CA, Champney M, Jensen RJ, Pitcher MR, Major PW. Correlation of skeletal maturation stages detemined by cervical vertebrae and hand-wrist evaluations. *Angle Orthod*. 2006;76(1):1–5.
10 Gandini P, Mancini M, Andreani F. A comparasion of handwrist bone and cervical vertebral analyses in measuring skeletal maturation. *Angle Orthod*. 2006;76(6):984–989.
11 Chartrand G, Cheng PM, Vorontsov E, et al. Deep learning: a primer for radiologists. *Radiographics*. 2017;37(7):2113–2131.
12 Jones LD, Golan D, Hanna SA, Ramachandran M. Artificial intelligence, machine learning and the evolution of healthcare. *Bone Joint Res*. 2017;7:223–225.
13 Amasya H, Yildirim D, Aydogan T, Kemaloglu N, Orhan K. Cervical vertebral maturation assessment on lateral cephalometric radiographs using artificial intelligence: comparison of machine learning classifier models. *Dentomaxillofacial Radiol*. 2020;49(5), 20190441. https://doi.org/10.1259/dmfr.20190441.
14 Moher D, Liberati A, Tetzlaff J, Altman DG, Group TP. Preferred reporting Items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009 Jul 21;6(7), e1000097.
15 Pai M, McCulloch M, Gorman JD, et al. Systematic reviews and meta-analyses: an illustrated, step-by-step guide. *Natl Med J India*. 2004;17(2):86–95.
16 Moola S, Munn Z, Tufanaru C, et al. Chapter 7: systematic reviews of etiology and risk. In: Aromataris E, ed. *Munn Z. Joanna Briggs Institute Reviewer's Manual*. The Joanna Briggs Institute; 2017.
17 Kök H, Acilar AM, İzgi MS. Usage and comparison of artificial intelligence algorithms for determination of growth and development by cervical vertebrae stages in orthodontics. *Prog Orthod*. 2019;20(1):41. https://doi.org/10.1186/s40510-019-0295-8.
18 Makaremi M, Lacaule C, Mohammad-Djafari A. Deep learning and artificial intelligence for the determination of the cervical vertebra maturation degree from lateral radiography. *Entropy*. 2019;21(12):1222.
19 Mohammad-Rahimi H, Motamadian SR, Nadimi M, et al. Deep learning for the classification of cervical maturation degree and pubertal growth spurts: a pilot study. *Korean J. Orthod.* 2022 Mar 25;52(2):112–122. https://doi.org/10.4041/kjod.2022.52.2.112. PMID: 35321950; PMCID: PMC8964471.
20 Seo H, Hwang JJ, Jung YH, Lee E, Nam OH, Shin J. Deep focus approach for accurate bone age estimation from lateral cephalogram. *J Dent Sci*. 2023 Jan;18(1):34–43. https://doi.org/10.1016/j.jds.2022.07.018. Epub 2022 Aug 20. PMID: 36643224; PMCID: PMC9831852.
21 Liao N, Dai J, Tang Y, Zhong Q, Mo S. iCVM: an interpretable deep learning model for CVM assessment under label uncertainty. *IEEE J. Biomed. Health Inf.* 2022 Aug;26(8):4325–4334. https://doi.org/10.1109/JBHI.2022.3179619. Epub 2022 Aug 11. PMID: 35653451.
22 Atici SF, Ansari R, Allareddy V, Suhaym O, Cetin AE, Elnagar MH. Fully automated determination of the cervical vertebrae maturation stages using deep learning with directional filters. *PLoS One*. 2022 Jul 1;17(7), e0269198. https://doi.org/10.1371/journal.pone.0269198. PMID: 35776715; PMCID: PMC9249196.
23 Kim EG, Oh IS, So JE, et al. Estimating cervical vertebral maturation with a lateral cephalogram using the convolutional neural network. *J. Clin. Med.* 2021 Nov 19;10(22):5400. https://doi.org/10.3390/jcm10225400. PMID: 34830682; PMCID: PMC8620598.
24 Kök H, Izgi MS, Acilar AM. Determination of growth and development periods in orthodontics with artificial neural network. *Orthod Craniofac Res*. 2021 Dec;24(Suppl 2):76–83. https://doi.org/10.1111/ocr.12443. Epub 2020 Dec 28. PMID: 33232582.
25 Zhou J, Zhou H, Pu L, et al. Development of an artificial intelligence system for the automatic evaluation of cervical vertebral maturation status. *Diagnostics*. 2021 Nov 25;11(12):2200. https://doi.org/10.3390/diagnostics11122200. PMID: 34943436; PMCID: PMC8700528.
26 Seo H, Hwang J, Jeong T, Shin J. Comparison of deep learning models for cervical vertebral maturation stage classification on lateral cephalometric radiographs. *J Clin Med*. 2021 Aug 15;10(16):3591. https://doi.org/10.3390/jcm10163591. PMID: 34441887; PMCID: PMC8397111.
27 Kok H, Izgi M, Acilar A. Evaluation of the artificial neural network and naive Bayes models trained with vertebra ratios for growth and development determination. *Turkish J Orthod*. 2021;34(1):2–9.
28 Amasya H, Cesur E, Yıldırım D, Orhan K. Validation of cervical vertebral maturation stages: artificial intelligence vs human observer visual analysis. *Am J Orthod Dentofacial Orthop*. 2020 Dec;158(6):e173–e179.
29 Fradkov AL. Early history of machine learning. *IFAC-PapersOnLine*. 2020;53(2):1385–1390.
30 Xie L, Tang W, Izadikhah I, et al. Intelligent quantitative assessment of skeletal maturation based on multi-stage model: a retrospective cone-beam CT study of cervical vertebrae. *Oral Radiol*. 2022 Jul;38(3):378–388. https://doi.org/10.1007/s11282-021-00566-y. Epub 2021 Sep 23. PMID: 34554389.
31 Feng X, Lu S, Li Y, Lin J. Establishment of an intelligent cervical vertebrae maturity assessment system based on cone beam CT data. *Zhejiang Da Xue Xue Bao Yi Xue Ban*. 2021 Apr 25;50(2):187–194. https://doi.org/10.3724/zdxbyxb-2021-0131. PMID: 34137225; PMCID: PMC8710263.
32 Kim DW, Kim J, Kim T, et al. Prediction of hand-wrist maturation stages based on cervical vertebrae images using artificial intelligence. *Orthod Craniofac Res*. 2021 Dec;24(Suppl 2):68–75. https://doi.org/10.1111/ocr.12514. Epub 2021 Aug 17. PMID: 34405944.
33 Aguiar LB, Caldas Mde P, Haiter Neto F, Ambrosano GM. A methodology to measure cervical vertebral bone maturation in a sample from low-income children. *Braz Dent J*. 2013;24(1):30–34.
34 Banumathi A, Raju S, Abhaikumar V. Diagnosis of dental deformities in cephalometry images using support vector machine. *J Med Syst*. 2011 Feb;35(1):113–119.
35 Litsas G, Athanasiou AE, Papadopoulos MA, et al. Dental calcification stages as determinants of the peak growth period. *J Orofac Orthop*. 2016;77:341–349. https://doi.org/10.1007/s00056-016-0040-6.
36 Moon JH, Hwang HW, Yu Y, Kim MG, Donatelli RE, Lee SJ. How much deep learning is enough for automatic identification to be reliable? *Angle Orthod*. 2020 Nov 1;90(6):823–830.
37 Santiago RC, Cunha AR, Júnior GC, et al. New software for cervical vertebral geometry assessment and its relationship to skeletal maturation—a pilot study. *Dentomaxillofacial Radiol*. 2014;43, 20130238.
38 Chen L, Lan Z, Xu X, Lin J, Hu H. Accuracy and repeatability of computer aided cervical vertebra landmarking in cephalogram. *J. Huazhong Univ. Sci. Technol. Med. Sci.* 2012 Feb;32(1):119–123.
39 Lindner C, Wang CW, Huang CT, Li CH, Chang SW, Cootes TF. Fully automatic system for accurate localisation and analysis of cephalometric landmarks in lateral cephalograms. *Sci Rep*. 2016 Sep 20;6, 33581.
40 Baptista RS, Quaglio CL, Mourad LM, et al. A semi-automated method for bone age assessment using cervical vertebral maturation. *Angle Orthod*. 2012 Jul;82(4):658–662.
41 Larson DB, Chen MC, Lungren MP, Halabi SS, Stence NV, Langlotz CP. Performance of a deep-learning neural network model in assessing skeletal maturity on pediatric hand radiographs. *Radiology*. 2017 Nov 2;287(1):313–322.
42 Mansourvar M, Ismail MA, Herawan T, Gopal Raj R, Abdul Kareem S, Nasaruddin FH. *Automated Bone Age Assessment: Motivation, Taxonomies, and Challenges [Internet]*. Comput Math Methods Med; 2013.
43 Burkov A. *The Hundred-Page Machine Learning Book*. Andriy Burkov; 2019.

44 Ranschaert ER, Morozov S, Algra PR. *Artificial Intelligence in Medical Imaging. Opportunities, Applications and Risks*. Cham: Springer Nature Switzerland AG; 2018.
45 Sokic E, Tiro A, Sokic Begovic E, Nakas E. Semi-Automatic assessment of cervical vertebral maturation stages using Cephalograph images and Centroid-based clustering. *Acta Stomatol Croat*. 2012;46:280–290.
46 Padalino S, Sfondrini MF, Chenuil L, Scudeller L, Gandini P. Reliability of skeletal maturity analysis using the cervical vertebrae maturation method on dedicated software. *Int Orthod*. 2014;12:483–493.

Update

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Erratum regarding missing Declaration of competing interest statements in previously published articles

Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of Journal of Oral Biology and Craniofacial Research.

The appropriate Declaration/Competing Interest statements, provided by the Authors, are included below.

16. "Cephalometric evaluation of soft tissue changes following Anterior Maxillary Osteotomy setback in Southern Dravidian population" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 1, January–February 2023, Pages 4–7] https://doi.org/10.1016/j.jobcr.2022.10.005

Declaration of competing interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

17. "Correlation of salivary visfatin levels in obese and NON-OBESE population with periodontal status" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 1, January–February 2023, Pages 67–70] https://doi.org/10.1016/j.jobcr.2022.11.004

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18. "Assessment of clinical outcome of surgically managed panfacial fractures with or without ancillary procedures. A 10-year retrospective study" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 79–83] https://doi.org/10.1016/j.jobcr.2022.12.001

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19. "Primary failure of eruption: From molecular diagnosis to therapeutic management" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 169–176] https://doi.org/10.1016/j.jobcr.2023.01.001

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20. "Paediatric condylar trauma – primary management considerations – A review" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 236–242] https://doi.org/10.1016/j.jobcr.2023.01.011

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21. "Comparative evaluation of hemispheric and masticatory laterality in complete dentures with two occlusal schemes – A prospective cohort study" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 272–276] https://doi.org/10.1016/j.jobcr.2023.02.005

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22. "Failure rates and factors associated with infrazygomatic crestal orthodontic implants - A prospective study" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 283–289] https://doi.org/10.1016/j.jobcr.2023.02.010

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24. "Effectiveness of photodynamic therapy (PDT) in the management of symptomatic oral lichen planus -A systematic review" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 353–359] https://doi.org/10.1016/j.jobcr.2023.03.003

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25. "Reconstruction of the mandibular condyle due to degenerative disease" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 2, March–April 2023, Pages 367–372] https://doi.org/10.1016/j.jobcr.2023.01.005

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26. "The influence of smoking on oral neutrophils and matrix metalloproteinase-8 in periodontitis patients before and after nonsurgical treatment" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 3, May–June 2023, Pages 442–447] https://doi.org/10.1016/j.jobcr.2023.05.004

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27. "Analysis of porosity, sealer dissolution and apical extrusion of endodontic sealers: A micro computed tomography study" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 4, July–August 2023, Pages 495–499] https://doi.org/10.1016/j.jobcr.2023.05.013

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28. "Importance of oral health in mental health disorders: An updated review" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 5, September–October 2023, Pages 544–552] https://doi.org/10.1016/j.jobcr.2023.06.003

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30. "Insertion of orthodontic temporary anchorage devices with free gingival grafting for phenotype modification of the peri-implant mucosa" [Journal of Oral Biology and Craniofacial Research, Volume 13, Issue 6, November–December 2023, Pages 727–730] https://doi.org/10.1016/j.jobcr.2023.09.005

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